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United States Patent	12419093
Kind Code	B2
Date of Patent	September 16, 2025
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Structure of flash memory cell

Abstract

A structure of flash memory cell includes a substrate. A floating gate is disposed on the substrate. A low dielectric constant (low-K) spacer is disposed on a sidewall of the floating gate. A trench isolation structure has a base part disposed in the substrate and a protruding part above the substrate protruding from the base part. The low-K spacer is sandwiched between the floating gate and the protruding part of the trench isolation structure.

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Appl. No.: 18/504165

Filed: November 08, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240072129 A1	Feb. 29, 2024

Related U.S. Application Data

continuation parent-doc US 16670870 20191031 US 11417734 20220816 child-doc US 17855700
division parent-doc US 17855700 20220630 US 11855156 child-doc US 18504165

Publication Classification

Int. Cl.: H01L21/762 (20060101); H10B41/30 (20230101); H10D30/68 (20250101); H10D64/01 (20250101); H10D64/66 (20250101); H01L21/3105 (20060101)

U.S. Cl.:

CPC **H10D64/035** (20250101); **H01L21/76224** (20130101); **H10B41/30** (20230201);
H10D30/68 (20250101); **H10D64/015** (20250101); **H10D64/679** (20250101);
H01L21/31051 (20130101)

Field of Classification Search

USPC: None

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a divisional application of and claims the priority benefit of a prior application Ser. No. 17/855,700, filed on Jun. 30, 2022. The prior application Ser. No. 17/855,700 is a continuation application of and claims the priority benefit of U.S. application Ser. No. 16/670,870, filed on Oct. 31, 2019. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference here and made a part of this specification.

BACKGROUND

1. Field of the Invention

(1) The present invention generally relates to semiconductor fabrication, and particularly to a structure of flash memory cell and method for fabricating the same.

2. Description of Related Art

(2) The non-volatile memory device is almost an absolutely required equipment in a digital electronic product. The digital electronic products such as computer system, mobile phone, camera, video apparatus, and so on, are also the necessary products in the daily life. Therefore, the non-volatile memory device is commonly required.

(3) The non-volatile memory device in an example is a flash memory, including a control gate and a floating gate. The floating gate is involved to store the charges according to the binary data as stored.

(4) When the device size is greatly reduced as intended, the gap between the cells is accordingly reduced. In accessing operation, the surrounding cells adjacent to the selected cell as currently accessed may be disturbed, then the cross-talk phenomenon between the target cell and the adjacent cells would occur. When the gap between the cells is greatly reduced, the parasitic capacitance between adjacent two FG's may get serious, and then the cross-talk effect gets serious. This effect would cause larger RC delay in operation.

(5) The cell structure of flash memory is still under development, so to expect the improvement of the whole performance of the memory device.

SUMMARY OF THE INVENTION

(6) The invention provides a structure of memory device. An isolation between the adjacent floating gates of the memory cells may be improved, while the gap between the floating gate in flash memory cells is reduced in fabrication trend.

- (7) In an embodiment, the invention provides a structure of flash memory cell includes a substrate. A floating gate is disposed on the substrate. A low dielectric constant (low-K) spacer is disposed on a sidewall of the floating gate. A trench isolation structure has a base part disposed in the substrate and a protruding part above the substrate protruding from the base part. The low-K spacer is sandwiched between the floating gate and the protruding part of the trench isolation structure.
- (8) In an embodiment, as to the structure of flash memory cell, the low-K spacer comprises a solid structure by the low-K dielectric.
- (9) In an embodiment, as to the structure of flash memory cell, the low-K spacer comprises an air spacer, having a free space filled with air.
- (10) In an embodiment, as to the structure of flash memory cell, it further includes a tunnel oxide layer between the substrate and the floating gate.
- (11) In an embodiment, as to the structure of flash memory cell, the tunnel oxide layer is disposed on the substrate between adjacent two of the base part of the trench isolation structure.
- (12) In an embodiment, as to the structure of flash memory cell, the protruding part of the trench isolation structure is narrower than a top of the base part.
- (13) In an embodiment, as to the structure of flash memory cell, the floating gate is extending over a portion of the base part of the trench isolation structure.
- (14) In an embodiment, as to the structure of flash memory cell, a top surface of low-K spacer is coplanar with a top surface of the floating gate.
- (15) In an embodiment, as to the structure of flash memory cell, a dielectric constant of the low-K spacer is smaller than a dielectric constant of the trench isolation structure.
- (16) In an embodiment, as to the structure of flash memory cell, the floating gate, the low-K spacer and the protruding part of the trench isolation structure form as a layer with uniform thickness.
- (17) In an embodiment, the invention provides a method for fabricating flash memory, including providing a substrate. A trench isolation structure is formed, having a base part disposed in the substrate and a protruding part above the substrate protruding from the base part. A dielectric spacer with low dielectric constant (low-K) is formed on a sidewall of the protruding part of the trench isolation structure. A floating gate is formed on the substrate abutting to the dielectric spacer. The dielectric spacer is sandwiched between the floating gate and the protruding part of the trench isolation structure.
- (18) In an embodiment, as to the method for fabricating flash memory, a dielectric constant of the dielectric spacer is smaller than a dielectric constant of the trench isolation structure.
- (19) In an embodiment, as to the method for fabricating flash memory, the method further includes removing the dielectric spacer to have a free space filled with air.
- (20) In an embodiment, as to the method for fabricating flash memory, the method further includes performing an oxidation process on the substrate before forming the floating gate to form a tunnel oxide layer between the substrate and the floating gate.
- (21) In an embodiment, as to the method for fabricating flash memory, the tunnel oxide layer is disposed on the substrate between adjacent two of the base part of the trench isolation structure.
- (22) In an embodiment, as to the method for fabricating flash memory, the step of forming the trench isolation structure comprising forming a plurality of dielectric islands on the substrate, wherein a gap is between adjacent two of the dielectric islands. Then a trench is formed in the substrate under the gap. Isolation material is disposed over the substrate to fill the trench and cover over the dielectric islands. The isolation material is polished to expose the dielectric islands, wherein the protruding part of the trench isolation structure is formed between the dielectric islands. The dielectric islands are removed.
- (23) In an embodiment, as to the method for fabricating flash memory, the floating gate is extending over a portion of the base part of the trench isolation structure.
- (24) In an embodiment, as to the method for fabricating flash memory, a top surface of dielectric spacer is coplanar with a top surface of the floating gate.

(25) In an embodiment, as to the method for fabricating flash memory, a dielectric constant of the dielectric spacer is smaller than a dielectric constant of silicon oxide.

(26) In an embodiment, as to the method for fabricating flash memory, the step of forming the floating gate comprising depositing a polysilicon layer over the substrate to at least cover over the protruding part of the trench isolation structure with the dielectric spacer. Then, the polysilicon layer is polished to expose the protruding part of the trench isolation structure with the dielectric spacer, wherein a residual part of the polysilicon layer provides the floating gate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings are included to provide a further understanding of the invention, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.
- (2) FIG. 1 is a drawing, schematically illustrating a mechanism of a cross-talk between memory cell during operation as looked into in the invention.
- (3) FIG. 2A to FIG. 2C are drawing, schematically illustrating process flows for fabricating a flash memory cell as looked into in the invention.
- (4) FIG. 3A to FIG. 3J are drawing, schematically illustrating process flows for fabricating a flash memory cell, according to an embodiment of the invention.
- (5) FIG. 4A to FIG. 4H are drawing, schematically illustrating process flows for fabricating a flash memory cell, according to an embodiment of the invention.

DESCRIPTION OF THE EMBODIMENTS

- (6) The invention is directed to a structure of flash memory cells and the method for fabricating the flash memory cells. The flash memory cells as fabricated may have less parasitic capacitance between floating gates. At least the cross talk effect between the target cell and the adjacent cells may at least be reduced.
- (7) The invention has looked into the performance between of the flash memory, which may be greatly reduced in size as the fabrication trend. In this situation, the distance between memory cells is accordingly reduced. When the operation voltages are applied to the memory cell as accessed, the adjacent cells may influence the target cell due to capacitance coupling with RC delay. The kind of cross-talk effect would reduce the performance of operation.
- (8) FIG. 1 is a drawing, schematically illustrating a mechanism of a cross-talk between memory cell during operation as looked into in the invention. Referring to FIG. 1, the invention has looked into the cross-talk effect in operation. During operation, a target memory cell, as indicated by round dot, is applied operation voltages at bit line (BL) and word line (WL). However, the other cells at the intersection nodes of the bit line (BL) and the word line (WL) not selected should be turned off by still applying a certain voltage on the bit line, in an example, so to assure the other cells are at OFF state.
- (9) However, the disturbed cells, as indicated by “x”, adjacent to the target cell are disturbed, because the word line voltage is still applied. As a result, cross-talk effect may be more serious.
- (10) To at least reduce the cross-talk effect, the invention has looked into the fabrication process for the flash memory cell to find the issue and then may provide the modification to reduce the issue, so to improve the performance of the operation.
- (11) The invention provides multiple embodiments to describe the inventions, However, the invention is not just limited to the embodiments. In addition, suitable combinations between the embodiments may also be made to have further embodiments.
- (12) FIG. 2A to FIG. 2C are drawing, schematically illustrating process flows for fabricating a

flash memory cell as looked into in the invention. Referring to FIG. 2A, a shallow trench isolation structure **102** has been formed in a substrate **100**. The shallow trench isolation structure **102** includes a base part disposed in the substrate **100** and a protruding part **104** above the substrate **100**, protruding from the base part. The protruding part **104** has the width **106** narrower than the top width of the base part, so that an indent region **108**, as preserved to form the floating gate, is formed.

(13) Referring to FIG. 2B, a tunnel oxide layer **110** is formed on the exposed surface of the substrate **100** within the indent region **108**. Referring to FIG. 2C, the floating gate **112**, such as polysilicon gate, is formed in the indent region **108**.

(14) As looked into in the invention, a gap between adjacent two floating gates **112** is determined by the width **106** of the protruding part of the shallow trench isolation structure **102**. The material of the shallow trench isolation structure **102** usually is silicon oxide with dielectric constant about 4.

(15) Since the gap between adjacent two floating gates **112** is greatly reduced in the cell structure to reduce the cell size, the parasitic capacitance between the floating gates **112** is getting large when the gap between the floating gates **112** is reduced. The capacitance coupling may exist and be not negligible.

(16) After at least looking into the issue above, the invention further provides embodiments. FIG. 3A to FIG. 3J are drawing, schematically illustrating process flows for fabricating a flash memory cell, according to an embodiment of the invention.

(17) Referring to FIG. 3A, a substrate **100** is provided as a structure base. Dielectric islands **40** are formed on the substrate **100**. The dielectric islands **40** are sacrificial material, such as silicon nitride, but the locations are preserved to form floating gates, which would replace the dielectric islands **40** in later fabrication process.

(18) Referring to FIG. 3B, the substrate **100** is etched to form a pattern of shallow trench **50**. As noted, the shallow trench **50** may extend into the bottom of the dielectric islands **40**. Referring to FIG. 3C, an oxide layer **60** is further disposed over the substrate **100** to fill into the shallow trench **50** and the cover the dielectric islands **40**. The dielectric islands **40** in an embodiment is nitride, different from the oxide layer **60**.

(19) Referring to FIG. 3D, a polishing process, such as chemical mechanical polishing (CMP) process, is performed over the substrate **100** to polish the oxide layer **60** until the dielectric islands **40** is exposed. The remaining portion of the oxide layer **60** forms a shallow trench isolation structure **200**.

(20) Referring to FIG. 3E, in an embodiment, the dielectric islands **40**, such as nitride, is removed by an etching process with an etching selection ratio to the nitride. Then the indent region **206** is formed on the substrate **100**. The indent region **206**, similar to FIG. 2A, is a region between the protruding parts **202** of the shallow trench isolation structure **200**. The protruding parts **202** of the shallow trench isolation structure **200** has a width **204** as viewed at the cross-section structure.

(21) Referring to FIG. 3F, a low dielectric constant (low-K) dielectric layer **208** is formed over the substrate **100**, being conformal with the protruding parts **202** of the shallow trench isolation structure **200** and the indent region **206**. The low-K dielectric layer **208** has dielectric constant less than 4 being relative to the silicon oxide. The thickness of the low-K dielectric layer **208** would determine how much the parasitic capacitance is reduced, as to be seen in subsequent processes. In other words, the thickness of the low-K dielectric layer **208** is preset.

(22) Referring to FIG. 3G, an etching back process is performed on the low-K dielectric layer **208**, so to form a low-K spacer **209** on sidewall of the protruding parts **202** of the shallow trench isolation structure **200**.

(23) Referring to FIG. 3H, a tunnel oxide layer **210** is formed on the exposed surface of the substrate **100** within the indent region **206**.

(24) Referring to FIG. 3I, a polysilicon layer **212** is disposed over the substrate **100** to fill the

indent region **206**.

(25) Referring to FIG. 3J, the polysilicon layer **212** is polished to expose the protruding parts **202** of the shallow trench isolation structure **200**. The remaining portion of the polysilicon layer **212** forms floating gates **216**.

(26) In this structure, the low-K spacer **209** is also disposed on a sidewall of the floating gate **216**. In other words, the low-K spacer **209** is sandwiched between the protruding parts **202** of the shallow trench isolation structure **200** and the floating gate **216**. The distance between the protruding parts **202** of the shallow trench isolation structure **200** may remain as preset. However, the gap **214** between the floating gates **216** may be increased because the width of the floating gates **216** is reduced due to low-K spacer **209**. As a result, the parasitic capacitance at the gap **214** between the floating gates **216** is reduced by a total effect from the low-K spacer **209**. The capacitance coupling between the floating gates **216** is reduced.

(27) FIG. 4A to FIG. 4H are drawing, schematically illustrating process flows for fabricating a flash memory cell, according to an embodiment of the invention. In a further embodiment, referring to FIG. 4A, the structure as fabricated in FIG. 3E is taken for further fabrication in another embodiment.

(28) Referring to FIG. 4B, a sacrificial layer **250**, such as nitride layer in different material from the oxide of the shallow trench isolation structure **200**, is formed over the substrate **100** by a conformal structure.

(29) Referring to FIG. 4C, an etching back process is performed on the sacrificial layer **250** to form a sacrificial spacer **252**.

(30) Referring to FIG. 4D, a tunnel oxide layer **210** is formed on the exposed surface of the substrate **100** within the indent region **206**.

(31) Referring to FIG. 4E, a polysilicon layer **212** is disposed over the substrate **100** to fill the indent region **206**.

(32) Referring to FIG. 4F, the polysilicon layer **212** is polished to expose the protruding parts **202** of the shallow trench isolation structure **200** and the sacrificial spacer **252**. The remaining portion of the polysilicon layer **212** forms floating gates **216**.

(33) Referring to FIG. 4G, the sacrificial spacer **252** is removed by etching process in an embodiment, in which an etching selection ratio to the material of the sacrificial spacer **252**, such as nitride, is set by a proper etchant. After the sacrificial spacer **252** is removed, a side space **254** is formed, serving as an air spacer, which has dielectric constant, rather close to 1.

(34) Referring to FIG. 4H, another dielectric layer **256** is formed on the floating gates **216**, such as oxide/nitride/oxide (ONO) layer in an embodiment. As a result, the side spaces **254** are sealed as air spacers between the floating gates **216** and the protruding parts **202** of the shallow trench isolation structure **200**. The parasitic capacitance is further reduced because the dielectric constant of the air is quite close to 1 while the dielectric constant for silicon oxide is about 4.

(35) The invention has provided the flash memory cell, which has the low-K spacer on the sidewall of the floating gates to at least reduce the parasitic capacitance. The low-K spacer has dielectric constant less than silicon oxide. The low-K spacer may be the air medium with dielectric constant rather close to 1. The method for fabricating the flash memory cell, in an embodiment, is also provided.

(36) It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present invention without departing from the scope or spirit of the invention. In view of the foregoing, it is intended that the present invention cover modifications and variations of this invention provided they fall within the scope of the following claims and their equivalents.

Claims

1. A structure of flash memory cell, comprising: a substrate; a floating gate, disposed on the substrate; a low dielectric constant (low-K) spacer, disposed on a sidewall of the floating gate; a trench isolation structure, having a base part disposed in the substrate and a protruding part above the substrate protruding from the base part; and a tunnel oxide layer between the substrate and the floating gate, wherein the low-K spacer is sandwiched between the floating gate and the protruding part of the trench isolation structure, wherein a peripheral bottom flat surface of the floating gate directly covers on a portion of the base part of the trench isolation structure, wherein the low-K spacer and the tunnel oxide layer are separated by a portion of the floating gate on the base part of the trench isolation structure, wherein the low-K spacer comprises an air spacer, having a free space filled with air and a dielectric constant being close to 1, wherein a dielectric constant of the low-K spacer is smaller than a dielectric constant of the trench isolation structure, wherein a bottom surface of the tunnel oxide layer is lower than a bottommost surface of the low-K spacer.
 2. The structure of flash memory cell in claim 1, wherein the tunnel oxide layer is disposed on the substrate between adjacent two of the base part of the trench isolation structure.
 3. The structure of flash memory cell in claim 1, wherein the protruding part of the trench isolation structure is narrower than a top of the base part.
 4. The structure of flash memory cell in claim 1, wherein a top surface of the low-K spacer is coplanar with a top surface of the floating gate.
 5. The structure of flash memory cell in claim 1, wherein the floating gate, the low-K spacer and the protruding part of the trench isolation structure are same in thickness and height in a normal direction of the substrate.
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